ANSWERS

I. 1. a) 2. c) 3. b) 4. d) 5. b)
6. d) 7. a) 8. c) 9. b) 10. a)
11. c) 12. b) 13. a) 14. d) 15. c)
16. b) 17. c) 18. a) 19. b) 20. a)

II. 25) $r_{XY} = 8$

26) $SD(Y) = 3.75$

III. 34) $r_{XY} = -0.95$

35) $\overline{X} = 13$, $\overline{Y} = 17$

36) when $Y = 50$, $\hat{X} = 62.8$

37) Regression equation X on Y: $\hat{X} = Y - 2$
Regression equation Y on X: $\hat{Y} = X + 2$
when $X = 3.5$, $\hat{Y} = 5.5$

IV. 41) (1) If X increases by 1 unit then Y increases by 3 units
(2) If X increases by 3 units then Y increases by 9 units

42) (1) Regression equation of X on Y is $X = Y + 6$; when $Y = 16$, $X = 22$
(2) Regression equation of Y on X is $Y = 0.89\ X - 2.59$
(3) $b_{xy} = 1$, $b_{yx} = 0.87$, $r = 0.93$

43) $Y = 0.45\ X + 111.53$, $Y = 138.53$ when age is 60 years.

44) $a = 16.4$, $b = -1.3$
Regression equation of X on Y is : $X = 16.4 - 1.3Y$

45) $X = 86$ when $Y = 60$

46) $Y = 0.32\ X + 21.84$

47) (a) Mean of $X = 13$ and mean of $Y = 17$. (b) $r = 0.6$

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